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Dr. Juliane Liepe
Facility for Data Sciences and Biostatistics
Max Planck Institute for Multidisciplinary Sciences
Am Faßberg 11, 37077 Göttingen, Germany

Application for Scientific IT Specialist / Research Software Engineer (Job Code 15-26)

Dear Dr. Liepe,

It is with great excitement and interest that I am writing this letter to apply for the Scientific IT Specialist / Research Software Engineer position (Job Code 15-26). The focus of the Facility for Data Sciences and Biostatistics on creating robust computational environments and supporting diverse research teams aligns perfectly to my background and career trajectory.

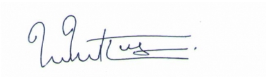
After my Ph.D. in astrochemistry and my postdoctoral work in marine ecosystem modeling, I transitioned to research software engineering and reproducible data pipelines. During my recent internship at HealthTwiSt GmbH, I gained extensive practical experience in clinical data engineering using R, Python, Bash and Linux. My core work there involves refactoring the monolithic DeGIR clinical data pipeline (tidyverse), reducing its codebase from 1,834 to 1,349 lines via config-driven ETL, and guaranteeing byte-identical outputs. Utilizing Git and GitHub extensively for version control, this experience combined with passing `devtools::check()` verifications enabled me to securely package the logic in to the `degirtools` R package. This work was a wonderful opportunity to learn and build highly reproducible data workflows requested by your open position.

My previous academic work also provided me with strong foundation in high-performance scientific computing. Working on theoretical modeling in my postdoc, I played a leading role in translating a legacy Fortran model into a high-performance Python simulation engine. Also, being a Fortran enthusiast and realizing the immense power of Python, specifically leveraging JAX vectorization for TPU and GPU parallelization, gives me a deep appreciation for modernizing scientific code. This was further solidified by specialized HPC training at the Jülich Supercomputing Centre (JSC). I am very comfortable working in Linux environments, where I work daily, and process large NetCDF datasets on HPC clusters and utilize Docker concepts. Recently, I also successfully built an automated Nextflow (DSL2) reproducible pipeline for Hepatitis Delta Virus (HDV) utilizing MAFFT, trimAl and Conda dependencies.

I am particularly drawn to your mission of acting as a multiplier for other scientists' productivity. For me, I have a strong willingness to learn and assimilate to help researchers transition between technologies. At HealthTwiSt GmbH, I authored `tidymodels` teaching materials on GitHub (via Quarto) to help researchers migrate from the legacy `caret` framework. This builds naturally upon my years of practical mentoring experience as a physics tutor and chess coach at Tutorwaves from 2018 to 2021. I am eager to apply my technical skills and team spirit to assist researchers and provide on-site training.

I am deeply motivated by the Max Planck Institute for Multidisciplinary Sciences' mission, bringing computational expertise in to an interdisciplinary environment bridging biology, chemistry, physics, and medicine across 1,000 employees from 66 nationalities. I am specifically eager to support your Facility. As my wife is currently completing her postdoctoral research at DESY in Hamburg, we are actively looking to relocate south as a dual-academic household. I am immediately available, highly motivated to establish our new base in Göttingen, and I am available in speaking with you for a personal interview.

Sincerely,



Dr. Thejus Mahajan

Enclosures: Curriculum Vitae, Degrees and Certificates